

Baer Invariants and the Birkhoff–Witt Theorem

P. J. Higgins

Department of Mathematics, King’s College, Strand, London W.C. 2

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1. Introduction

The Birkhoff–Witt theorem asserts that, under certain conditions on a Lie algebra M (over a commutative ring R), the graded algebra associated with the enveloping algebra of M is canonically isomorphic with the symmetric algebra of the underlying module of M . It then follows that M is embedded in its enveloping algebra. It is known that not all Lie algebras are so embedded, but the theorem has been proved in the following cases:

- (i) when M is a free R -module, for any R (Birkhoff [1], Witt [7]);
- (ii) when R is a Dedekind domain, for any M (Lazard [5], Cartier [2]);
- (iii) when R is an algebra over the rationals, for any M (Cohn [3]).

These conditions refer only to the module structure of M , and we aim to give a unified treatment in which the Lie product on M and the enveloping algebra of M are eliminated from the discussion at an early stage. They are replaced by a homological invariant $B(M)$, defined for an arbitrary module M , which depends only on the tensor algebra of M and is one of the Baer invariants defined for algebras by Fröhlich [4]. The vanishing of $B(M)$ implies the Birkhoff–Witt theorem for all Lie algebras with underlying module M , and we shall show that $B(M) = 0$ in the three cases mentioned above. In fact the condition $B(M) = 0$ is necessary and sufficient for the validity of a closely related embedding theorem for “Lie structures over M ” in “associative structures over M .” The definition of these concepts is our starting-point.

2. Lie Structures and Associative Structures

Let R be a commutative ring (with identity) and let M be an R -module. We denote by $T(M) = \bigoplus_{n \geq 0} (\otimes^n M)$ the tensor algebra of M and identify M with the homogeneous part of $T(M)$ of degree 1. (All tensor products are taken over R unless otherwise stated). The canonical map from $T(M)$ to the symmetric algebra $S(M)$ has as kernel the ideal of $T(M)$ generated by all commutators $xy - yx$ ($x, y \in M$). We denote this kernel by $K(M)$; it is a homogeneous ideal of $T(M)$.

If we consider $K(M)$ as a $T(M)$ -bimodule, it is generated by the commutators $xy - yx$, and these satisfy some obvious relations which we use as axioms in the following definition. A *Lie structure over the R -module M* is a $T(M)$ -bimodule A together with a bilinear function $M \otimes M \rightarrow A$ (denoted by $x \otimes y \mapsto \langle x, y \rangle$), satisfying the axioms:

$$(L1) \quad \langle x, x \rangle = 0 \quad (x \in M);$$

$$(L2) \quad \langle x, y \rangle t(uv - vu) = (xy - yx)t\langle u, v \rangle \quad (x, y, u, v \in M, t \in T(M));$$

$$(L3) \quad (\langle x, y \rangle z - z\langle x, y \rangle) + (\langle y, z \rangle x - x\langle y, z \rangle) + (\langle z, x \rangle y - y\langle z, x \rangle) = 0 \quad (x, y, z \in M).$$

It is easy to check that $K(M)$ is a Lie structure with $\langle x, y \rangle = xy - yx$. So is $T(M)$ with the same definition of $\langle x, y \rangle$, and other examples will occur below.

As with Lie algebras, one can easily obtain Lie structures from similar associative structures. We define an *associative structure over M* to be a $T(M)$ -bimodule B together with a bilinear function $M \otimes M \rightarrow B$ (denoted by $x \otimes y \mapsto (x, y)$) satisfying the associative law:

$$(A) \quad (x, y)z = x(y, z) \quad (x, y, z \in M).$$

Then B becomes a Lie structure over M if we define $\langle x, y \rangle = (x, y) - (y, x)$. Axioms (L1) and (L3) are obviously satisfied. Axiom (L2) need only be checked when $t = z_1 z_2 \cdots z_n$ ($z_i \in M$), and in this case axiom (A) implies $(x, y)z_1 z_2 \cdots z_n uv = xy z_1 z_2 \cdots z_n (u, v)$, from which (L2) follows easily.

For any given module M there is a universal Lie structure $L(M)$ over M which can be described as follows. $L(M)$ is generated as $T(M)$ -bimodule by symbols $\langle x, y \rangle$, one for each pair of elements x, y of M , and has defining relations (L1), (L2), (L3) together with the relations that assert the bilinearity of the function $\langle \cdot, \cdot \rangle$. This $L(M)$ is characterised up to isomorphism by the universal property: if $(A, \langle \cdot, \cdot \rangle)$ is any Lie structure over M then there is a unique morphism $L(M) \rightarrow A$ of $T(M)$ -bimodules such that $\langle x, y \rangle \mapsto \langle \langle x, y \rangle \rangle$ for all $x, y \in M$.

There is also, of course, a universal associative structure over M , but we do not need a special notation for it since it can easily be identified. In $T(M)$, the ideal $M^2 T(M) = \bigoplus_{n \geq 2} (\otimes^n M)$ is an associative structure over M with $(x, y) = xy$, and we have

Theorem 1. $M^2 T(M)$ is the universal associative structure over M .

Proof. Let $(B, (,))$ be any associative structure over M . For $j \geq 2$ we can map $\otimes^j M$ to B by the rule $x_1 \otimes x_2 \otimes \cdots \otimes x_j \mapsto (x_1, x_2)x_3 \cdots x_j$ since the latter is an R -multilinear function. This gives a map $\theta : M^2 T(M) \rightarrow B$ which is clearly a homomorphism of right $T(M)$ -modules and maps xy to (x, y) . To show that it is also a left $T(M)$ -homomorphism it is enough to

show that $y_1 y_2 \cdots y_n (x_1, x_2) x_3 \cdots x_j = (y_1, y_2) y_3 \cdots y_n x_1 \cdots x_j$, ($y_r, x_s \in M$), and this is a consequence of the associative law for B . The map θ is unique because the elements xy ($x, y \in M$) generate $M^2 T(M)$ as $T(M)$ -bimodule. $\square$

It is unfortunately not always true that $K(M)$ is the universal Lie structure over M . Indeed, it is precisely when $K(M) \cong L(M)$ that we can prove the Birkhoff–Witt theorem for Lie algebras on the module M . The proof of this theorem given in the next section is essentially that of Lazard [5], with some simplifications made possible by the axiomatic approach.

3. The Birkhoff–Witt Theorem

Suppose that we are given a multiplication $M \otimes M \rightarrow M$ ($x \otimes y \mapsto [x, y]$) which makes M a Lie algebra over R . The elements $\langle x, y \rangle = xy - yx - [x, y]$ of $T = T(M)$ ($x, y \in M$) generate a (non-homogeneous) ideal J , and the quotient algebra $E = T/J$ is the *enveloping algebra* of the Lie algebra M . The filtration $T_0 \subset T_1 \subset T_2 \subset \cdots$ of T given by $T_n = \bigoplus_{i \leq n} (\otimes^i M)$ induces filtrations of $K = K(M)$, $S = S(M)$, J and E as follows: $K_n = K \cap T_n$, $S_n = (T_n + K)/K$, $J_n = J \cap T_n$, $E_n = (T_n + J)/J$. E is then a filtered algebra and we denote by $G = \bigoplus_{n \geq 0} (E_n/E_{n-1})$ the associated graded algebra¹. The homogeneous components of G are

$$E_n/E_{n-1} \cong (T_n + J)/(T_{n-1} + J) \cong T_n/(T_n \cap (T_{n-1} + J)) \cong T_n/(T_{n-1} + J_n).$$

Now S is itself graded with homogeneous components

$$S_n/S_{n-1} \cong T_n/(T_{n-1} + K_n),$$

and it is clear from the definition of J that $T_{n-1} + J_n \supset T_{n-1} + K_n$. We therefore have canonical surjections $\sigma_n : S_n/S_{n-1} \rightarrow E_n/E_{n-1}$ with kernels $(T_{n-1} + J_n)/(T_{n-1} + K_n)$. These give the canonical surjection $\sigma : S \rightarrow G$ which is in fact the algebra homomorphism induced by the canonical map from M to the (commutative) algebra G . If σ is an isomorphism we say that the Lie algebra M has the *Birkhoff–Witt property*, and this is clearly equivalent to the condition

$$J_n \subset T_{n-1} + K_n \quad (n \geq 1). \quad (1)$$

Theorem 2. *If $(K(M), xy - yx)$ is the universal Lie structure over M then every Lie algebra on M has the Birkhoff–Witt property.*

Proof. Following Lazard, we introduce modules $J_{(n)}$ ($n \geq 1$) which consist of those elements of J that are most obviously in T_n , namely

$$J_{(n)} = \sum (T_r \langle x, y \rangle T_s \mid x, y \in M, r + s = n - 2).$$

Then $J_{(1)} = 0$, $J_{(2)}$ is the R -module spanned by all $\langle x, y \rangle = xy - yx - [x, y]$, and $J_{(1)} \subset J_{(2)} \subset \cdots \subset J_{(n)} \subset \cdots \subset J$. Also $J_{(n)} \subset J_n$, and $\bigcup_n J_{(n)} = J$. We write $A_n = J_{(n)}/J_{(n-1)}$ ($n \geq 2$), and form the graded R -module $A = \bigoplus_{n \geq 2} A_n$ associated with this new filtration of J . Since $J_{(n)}M + MJ_{(n)} \subset J_{(n+1)}$, A has the structure of a T -bimodule with $A_n M + M A_n \subset A_{n+1}$. Also, $\langle x, y \rangle \in J_{(2)} = A_2$ for all $x, y \in M$.

¹ *Comment by DG:* The direct sum ranged over $n \geq 1$ in the original.

Lemma. $(A, \langle, \rangle)$ is a Lie structure over M .

Proof. (L1): this is trivially true since $\langle x, x \rangle = [x, x] = 0$ in M .

(L2): let $w = \langle x, y \rangle t(uv - vu) - (xy - yx)t\langle u, v \rangle$, where $x, y, u, v \in M$, and where $t \in T$ is homogeneous of degree n . Then, calculating in A , we have $w \in A_{n+4}$ since $\langle x, y \rangle \in A_2$ and $\langle u, v \rangle \in A_2$. But if we calculate in $J_{(n+4)}$ we find that

$$w = \langle x, y \rangle t\{\langle u, v \rangle + [u, v]\} - \{\langle x, y \rangle + [x, y]\} t\langle u, v \rangle = \langle x, y \rangle t[u, v] - [x, y] t\langle u, v \rangle,$$

and this lies in $J_{(n+3)}$ since $[u, v]$ and $[x, y]$ lie in M . Hence $w = 0$ in A , and (L2) follows for all t by linearity.

(L3): let $x, y, z \in M$, and put

$$u = \{\langle x, y \rangle z - z\langle x, y \rangle\} + \{\langle y, z \rangle x - x\langle y, z \rangle\} + \{\langle z, x \rangle y - y\langle z, x \rangle\}.$$

Then, calculating in A , we have $u \in A_3$. On the other hand, calculating in $J_{(3)}$, if we replace $\langle x, y \rangle$ by $xy - yx - [x, y]$, we obtain

$$u = \{[x, y]z - z[x, y]\} + \{[y, z]x - x[y, z]\} + \{[z, x]y - y[z, x]\},$$

the other terms cancelling. Now $[x, y] \in M$, so $\langle [x, y], z \rangle \in J_{(2)}$, that is,

$$[x, y]z - z[x, y] \equiv [[x, y], z] \pmod{J_{(2)}}.$$

Permuting x, y, z cyclically and adding, we therefore have $u \in J_{(2)}$ by the Jacobi law in M , and this means that $u = 0$ in A . The lemma is now proved. $\square$

To prove the theorem we use the hypothesis that $K(M)$ is the *universal* Lie structure over M to obtain a morphism $\theta : K \rightarrow A$ of T -bimodules sending $xy - yx$ to $\langle x, y \rangle$ for all $x, y \in M$. We claim that θ is an isomorphism. For let δ_n denote the R -linear map which sends any element of T to its homogeneous part of degree n . It is clear from the definition of $J_{(n)}$ that δ_n maps $J_{(n)}$ into K_n and therefore induces a map

$$\delta_n^* : A_n = J_{(n)}/J_{(n-1)} \rightarrow K_n/K_{n-1}.$$

The maps δ_n^* combine to give a map $\delta : A \rightarrow \bigoplus (K_n/K_{n-1}) = K$ which sends $\langle x, y \rangle$ to $xy - yx$ ($x, y \in M$), and it is easy to check that δ is a morphism of T -bimodules. Since A and K are generated as T -bimodules by all $\langle x, y \rangle$ and all $xy - yx$, respectively, we see that θ and δ are inverse isomorphisms. In particular, $\delta_n^* : J_{(n)}/J_{(n-1)} \rightarrow K_n/K_{n-1}$ is an injection and it follows that every element of $J_{(n)}$ not in $J_{(n-1)}$ has leading term of degree exactly n . Since $\bigcup_n J_{(n)} = J$, this implies that $J_{(n)} = J \cap T_n = J_n$ for all n ². But $J_{(n)} \subset K_n + T_{n-1}$, so we have established the condition (1) which is equivalent to the Birkhoff–Witt property. $\square$

² *Comment by DG:* Here are the details of this argument: We must show that $J_{(n)} = J_n$. Since $J_{(n)} \subset J_n$ is known, it suffices to show that $J_n \subset J_{(n)}$. So let $t \in J_n$. Then, $t \in J$, so that $t \in J_{(m)}$ for some $m \geq 0$. Consider the smallest such m . Then, $t \in J_{(m)} \setminus J_{(m-1)}$, so that the residue class of t in $J_{(m)}/J_{(m-1)}$ is nonzero. Since $\delta_m^* : J_{(m)}/J_{(m-1)} \rightarrow K_m/K_{m-1}$ is an injection (as we showed above), we conclude that $\delta_m(t)$ is nonzero as well. In other words, t has a nonzero homogeneous part of degree m . Since $t \in J_n \subset T_n$, this entails $m \leq n$, so that $t \in J_{(m)} \subset J_{(n)}$. Having proved this for each $t \in J_n$, we thus obtain $J_n \subset J_{(n)}$, just as desired.

4. Lie Structures over Free Modules

Before investigating Lie structures over arbitrary modules M we need to know the situation for the special case when M is free. Our next theorem, combined with Theorem 2, gives a new proof of the Birkhoff–Witt theorem in this case.

Theorem 3. *Let M be a free R -module. Then every Lie structure $(A, \langle, \rangle)$ over M can be embedded in an associative structure $(B, (,))$ over M so that $\langle x, y \rangle = (x, y) - (y, x)$.*

Proof. Let A be a given Lie structure and put $B = A \oplus S(M)$. We shall show how to make B an associative structure with the required property. Let X be a basis for M over R and take a fixed total ordering $\leq$ of X . If $x_i \in X$ we denote by ξ_i its image in $S(M) = S$. Then S has a basis consisting of all products $\xi_1 \xi_2 \cdots \xi_n$ ($n \geq 0$), where the x_i are in X and $x_1 \leq x_2 \leq \cdots \leq x_n$.

We make B a T -bimodule as follows. The action of T on A is to be the given action. To define the action of T on S we need only define maps $M \otimes S \rightarrow B$ ($m \otimes \sigma \rightarrow m\sigma$) and $S \otimes M \rightarrow B$ ($\sigma \otimes m \rightarrow \sigma m$) such that $(m\sigma)n = m(\sigma n)$ for all $m, n \in M$, $\sigma \in S$. Since M is free we can define $x\sigma$ and σy arbitrarily in B for $x, y \in X$ and σ a basis element $\xi_1 \xi_2 \cdots \xi_n$ of S , and we need only check that

$$(x\sigma)y = x(\sigma y) \quad (2)$$

in this case. So let $\sigma = \xi_1 \xi_2 \cdots \xi_n$ ($x_1 \leq x_2 \leq \cdots \leq x_n$ in X) and let ξ, η be the images of x, y in S . We define

$$x\sigma = x \circ \sigma + \xi\sigma, \quad \sigma y = \sigma \circ y + \sigma\eta,$$

where $\xi\sigma, \sigma\eta$ are products in S and $x \circ \sigma, \sigma \circ y$ are the elements of A defined by

$$\begin{aligned} x \circ \sigma &= \sum_{x_i < x} x_1 x_2 \cdots x_{i-1} \langle x, x_i \rangle x_{i+1} \cdots x_n, \\ \sigma \circ y &= \sum_{x_i > y} x_1 x_2 \cdots x_{i-1} \langle x_i, y \rangle x_{i+1} \cdots x_n, \end{aligned}$$

with the convention that $x \circ 1 = 1 \circ y = 0$. (These definitions are just an imitation of the operations in the associative structure $T(M)$ in terms of the splitting $T(M) = K(M) + S(M)$). Since $(\xi\sigma)\eta = \xi(\sigma\eta)$ in S , equation (2) is equivalent to

$$(\xi\sigma) \circ y - x(\sigma \circ y) = x \circ (\sigma\eta) - (x \circ \sigma)y. \quad (3)$$

³ To simplify the notation, let $\tau = \xi\sigma\eta$ (product in S), and rename x, y and $x_1, x_2, \dots, x_n$ so that $\tau = \eta_1 \eta_2 \cdots \eta_k$ with⁴ $y_1 \leq y_2 \leq \cdots \leq y_k$ ($k = n + 2$) and $x = y_r, y = y_s$ ($r \neq s$). Then $\sigma = (\eta_1 \eta_2 \cdots \eta_k)_{\widehat{r}\widehat{s}}$, where the subscripts $\widehat{r}, \widehat{s}$ denote that the factors with subscripts

³ *Comment by DG:* The following proof of (3) is rather heavy on the reader's concentration. An alternative proof, found by GPT-5.5, is included in the Appendix below.

⁴ *Comment by DG:* The η_i are understood to be images of the y_i in $S(M) = S$.

r, s are to be omitted. Also $\xi\sigma = (\eta_1\eta_2\cdots\eta_k)_{\widehat{s}}$ and $\sigma\eta = (\eta_1\eta_2\cdots\eta_k)_{\widehat{r}}$. Writing⁵ $\epsilon_{pq} = 0$ if $p \leq q$ and $\epsilon_{pq} = 1$ if $p > q$, the left hand side of (3) becomes

$$\begin{aligned} (\xi\sigma) \circ y - x(\sigma \circ y) &= \sum_j \epsilon_{js}(y_1 \cdots y_{j-1} \langle y_j, y_s \rangle y_{j+1} \cdots y_k)_{\widehat{s}} \\ &\quad - y_r \sum_{j \neq r} \epsilon_{js}(y_1 \cdots y_{j-1} \langle y_j, y_s \rangle y_{j+1} \cdots y_k)_{\widehat{r}\widehat{s}}. \end{aligned}$$

The terms $j \neq r$ in the first sum appear in the second sum with y_r moved to the left hand end. This move can be accomplished by adding terms containing commutators $y_i y_r - y_r y_i$ and possibly $\langle y_j, y_s \rangle y_r - y_r \langle y_j, y_s \rangle$. We therefore have $(\xi\sigma) \circ y - x(\sigma \circ y) = U + V + W$, where

$$\begin{aligned} U &= \epsilon_{rs}(y_1 \cdots y_{r-1} \langle y_r, y_s \rangle y_{r+1} \cdots y_k)_{\widehat{s}}, \\ V &= - \sum_{j \neq r} \sum_{\substack{i \neq s \\ i \neq j}} \epsilon_{js} \epsilon_{ri}(y_1 \cdots y_{i-1} (y_r y_i - y_i y_r) y_{i+1} \cdots y_{j-1} \langle y_j, y_s \rangle y_{j+1} \cdots y_k)_{\widehat{r}\widehat{s}}, \\ W &= \sum_j \epsilon_{js} \epsilon_{rj}(y_1 \cdots y_{j-1} (\langle y_j, y_s \rangle y_r - y_r \langle y_j, y_s \rangle) y_{j+1} \cdots y_k)_{\widehat{r}\widehat{s}}. \end{aligned}$$

(The notation in V is not meant to imply that $i < j$). Similarly, the right hand side of (3) is $U' + V' + W'$, where

$$\begin{aligned} U' &= \epsilon_{rs}(y_1 \cdots y_{s-1} \langle y_r, y_s \rangle y_{s+1} \cdots y_k)_{\widehat{r}}, \\ V' &= - \sum_{i \neq s} \sum_{\substack{j \neq r \\ j \neq i}} \epsilon_{ri} \epsilon_{js}(y_1 \cdots y_{i-1} \langle y_r, y_i \rangle y_{i+1} \cdots y_{j-1} (y_j y_s - y_s y_j) y_{j+1} \cdots y_k)_{\widehat{r}\widehat{s}}, \\ W' &= \sum_j \epsilon_{rj} \epsilon_{js}(y_1 \cdots y_{j-1} (y_s \langle y_r, y_j \rangle - \langle y_r, y_j \rangle y_s) y_{j+1} \cdots y_k)_{\widehat{r}\widehat{s}}. \end{aligned}$$

Now $V = V'$ by axiom (L2) in A . Also, by axiom (L3),

$$\begin{aligned} W - W' &= \sum_{s < j < r} (y_1 \cdots y_{j-1} (y_j \langle y_s, y_r \rangle - \langle y_s, y_r \rangle y_j) y_{j+1} \cdots y_k)_{\widehat{r}\widehat{s}} \\ &= \epsilon_{rs} \{ (y_1 \cdots y_{r-1} \langle y_s, y_r \rangle y_{r+1} \cdots y_k)_{\widehat{s}} \\ &\quad - (y_1 \cdots y_{s-1} \langle y_s, y_r \rangle y_{s+1} \cdots y_k)_{\widehat{r}} \} \\ &= U' - U \end{aligned}$$

since (L1) implies that $\langle y_s, y_r \rangle = -\langle y_r, y_s \rangle$. Thus equation (3) is satisfied and B is a T -bimodule.

Now for $x, y \in X$ we have

$$x\eta = \xi y = \begin{cases} \xi\eta & \text{if } x \leq y \\ \xi\eta + \langle x, y \rangle & \text{if } x \geq y. \end{cases} \quad (4)$$

⁵ *Comment by DG:* The “ p ” and “ q ” in this notation were called “ r ” and “ s ” in the original. I renamed them to disambiguate them from the r and s already defined.

We may therefore define $(x, y) = x\eta = \xi y$ for $x, y \in X$, and extend linearly to the whole of M . Then B is an associative structure over M since the equation $(x, y)z = x(y, z)$ is R -multilinear and is a special case of (2) when $x, y, z \in X$. Finally, $(x, y) - (y, x) = x\eta - y\xi = \langle x, y \rangle$ by (4) when $x, y \in X$, and the equality holds for $x, y \in M$ by linearity. $\square$

Corollary 1. *If M is a free R -module then $K(M)$ is the universal Lie structure over M .*

Proof. Let $(A, \langle, \rangle)$ be any Lie structure over M and embed A in the associative structure B as above. By Theorem 1 there is a map $\theta : M^2 T(M) \rightarrow B$ of $T(M)$ -bimodules sending xy to (x, y) for $x, y \in M$. Since $(x, y) - (y, x) = \langle x, y \rangle \in A$, θ induces a map of $T(M)$ -bimodules $K(M) \rightarrow A$ sending $xy - yx$ to $\langle x, y \rangle$. $\square$

Using Theorem 2 we now obtain

Corollary 2. *Every Lie algebra over R whose underlying module is free has the Birkhoff–Witt property.*

5. Baer Invariants of Tensor Algebras

We now introduce two invariants of a module M analogous to the invariants $[F, F]/[F, R]$ and $(R \cap [F, F])/[F, R]$ for a group presented as a quotient F/R of a free group. In Fröhlich's notation [4] they are $D_0 V(T(M))$ and $D_1 U(T(M))$, where U and V are the functors on algebras associated with the variety of commutative algebras: $U(T(M)) = S(M)$, $V(T(M)) = K(M)$. We recall briefly their definition and main properties.

We start with a short exact sequence of R -modules

$$0 \rightarrow Q \rightarrow P \rightarrow M \rightarrow 0 \quad (5)$$

with P projective, and we make identifications so that $Q \subset P \subset T(P)$. Then we have a commutative diagram with exact rows and columns

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & K(P) \cap \overline{Q} & \longrightarrow & \overline{Q} & \longrightarrow & \overline{Q}^* \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & K(P) & \longrightarrow & T(P) & \longrightarrow & S(P) \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & K(M) & \longrightarrow & T(M) & \longrightarrow & S(M) \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & 0 & & 0 & & 0
 \end{array}$$

where $\overline{Q}$ is the ideal $T(P)QT(P)$ of $T(P)$ generated by Q , and $\overline{Q}^*$ is its image in $S(P)$ ⁶. Let W be the set of words

$$w_i(\mathbf{x}) = w_i(x_1, x_2, \dots, x_n) = x_1 x_2 \cdots x_{i-1} [x_i, x_{i+1}] x_{i+2} \cdots x_n$$

($n \geq 2, 1 \leq i \leq n-1$), where $[a, b]$ denotes, as always from now on, the additive commutator $ab - ba$. If the x_j take values running through P then W generates $K(P)$ as additive group. Let W' be the set of derived words $w'_i(\mathbf{x}, \mathbf{y}) = w_i(\mathbf{x} + \mathbf{y}) - w_i(\mathbf{x})$. If the x_j run through P and the y_j run through Q , W' generates (as additive group) an R -module Z . Clearly the image of any such element $w'_i(\mathbf{p}, \mathbf{q})$ in $T(M)$ or in $S(P)$ is 0, so $Z \subset K(P) \cap \overline{Q}$, and it is not difficult to check that in fact Z is the ideal of $T(P)$ generated by all $[t, q]$ with $t \in T(P)$

⁶ *Comment by DG:* The exactness of the rows is obvious. The exactness of the second and third columns follows from known results (Theorem 32 and Theorem 62 in Darij Grinberg, *A few classical results on tensor, symmetric and exterior powers*, 17 April 2026, to name just one reference). The exactness of the first column follows from the exactness of the second.

and $q \in \overline{Q}$ ⁷. We write

$$B(M) = (K(P) \cap \overline{Q})/Z \quad \text{and} \quad C(M) = K(P)/Z.$$

(These are respectively $D_1UT(M)$ and $D_0VT(M)$).

Now $K(P)$ and $\overline{Q}$ and Z are $T(P)$ -bimodules and satisfy⁸ $K(P)\overline{Q} + \overline{Q}K(P) \subset Z$ (because e.g. $p_1p_2 \cdots [p_i, p_{i+1}] \cdots q \cdots p_n$ is of the form $w'_i(\mathbf{p}, \mathbf{q})$). Hence $B(M)$ and $C(M)$ are $T(M)$ -bimodules, and we have an exact sequence of $T(M)$ -bimodules:

$$0 \rightarrow B(M) \rightarrow C(M) \rightarrow K(M) \rightarrow 0. \quad (6)$$

As the notation suggests, $B(M)$ and $C(M)$ depend only on M and not on its presentation (5). To see this directly, let $0 \rightarrow Q' \rightarrow P' \rightarrow M \rightarrow 0$ be another presentation of M with P'

⁷*Comment by DG:* In the original: “ $Z = [T(P), \overline{Q}]$, the ideal of $T(P)$ generated by all $[t, q]$ ($t \in T(P)$, $q \in \overline{Q}$)”. But the notation $[T(P), \overline{Q}]$ for this is rather nonstandard; usually it would mean just the R -linear span of all commutators $[t, q]$ with $t \in T(P)$ and $q \in \overline{Q}$, and this span does not equal Z .

Here is a brief outline of the proof: It is easy to see from the definition that

$$Z = \text{span}\{w_i(x_1, x_2, \dots, x_n) \mid \text{each } x_j \text{ belongs to } P, \text{ and at least one } x_j \text{ belongs to } Q\}.$$

(Indeed, this is perhaps the better definition of Z .) This shows that Z is an ideal of $T(P)$. Now, let $\tilde{Z}$ be the ideal of $T(P)$ generated by all $[t, q]$ with $t \in T(P)$ and $q \in \overline{Q}$. We must prove that $Z = \tilde{Z}$. We shall achieve this by proving both $\tilde{Z} \subseteq Z$ and $Z \subseteq \tilde{Z}$:

1. To prove $\tilde{Z} \subseteq Z$, we need to show that every $t \in T(P)$ and $q \in \overline{Q}$ satisfy $[t, q] \in Z$ (since Z is an ideal of $T(P)$). By linearity, we can assume that $t = x_1x_2 \cdots x_n$ with $x_1, x_2, \dots, x_n \in P$, and that $q = aq'b$ with $a = y_1y_2 \cdots y_k \in T(P)$ and $b = y_{k+1}y_{k+2} \cdots y_\ell \in T(P)$ and $q' \in Q$. But then we have

$$[t, q] = [t, aq'b] = [t, a]q'b + a[t, q']b + aq'[t, b]$$

and

$$[t, q'] = [x_1x_2 \cdots x_n, q'] = \sum_i x_1x_2 \cdots x_{i-1}[x_i, q']x_{i+1}x_{i+2} \cdots x_n.$$

Substituting the latter equality into the former, we obtain

$$[t, q] = [t, a]q'b + a \left(\sum_i x_1x_2 \cdots x_{i-1}[x_i, q']x_{i+1}x_{i+2} \cdots x_n \right) b + aq'[t, b].$$

All addends on the right hand side (once expanded) are of the form $w_i(x_1, x_2, \dots, x_n)$ where each x_j belongs to P and at least one x_j belongs to Q . Thus, they belong to Z . Hence, $[t, q] \in Z$. So we have shown that $\tilde{Z} \subseteq Z$.

2. In order to prove the reverse inclusion $Z \subseteq \tilde{Z}$, we must show that each $w_i(x_1, x_2, \dots, x_n)$ with all $x_j \in P$ and at least one $x_j \in Q$ belongs to $\tilde{Z}$. This is clear if the one x_j that belongs to Q is x_i or x_{i+1} , because in this case the $[x_i, x_{i+1}]$ bracket in the middle of $w_i(x_1, x_2, \dots, x_n)$ itself has the form $[t, q]$ with $t \in T(P)$ and $q \in \overline{Q}$ (actually, $q = x_j \in Q$). In all other cases, we first move this one x_j to a position either just left or just right of the $[x_i, x_{i+1}]$ bracket by successively commuting it past the other x_k 's (the commutators will lie in $\tilde{Z}$), and then use $x_j[x_i, x_{i+1}] = [x_jx_i, x_{i+1}] - [x_j, x_{i+1}]x_i$ (if x_j ended up left of the bracket) or $[x_i, x_{i+1}]x_j = [x_i, x_{i+1}x_j] - x_{i+1}[x_i, x_j]$ (if on the right), obtaining a right hand side belonging to $\tilde{Z}$ in both cases (since x_j , x_jx_i and $x_{i+1}x_j$ all belong to $\overline{Q}$).

⁸*Comment by DG:* Added “and Z ” as well as a few filler words (to avoid overfull hbox).

projective. Then there are R -linear maps σ, σ_* making the following diagram commute:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & Q & \longrightarrow & P & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow \sigma_* & & \downarrow \sigma & & \downarrow 1_M & & \\ 0 & \longrightarrow & Q' & \longrightarrow & P' & \longrightarrow & M & \longrightarrow & 0. \end{array}$$

The map $T(\sigma) : T(P) \rightarrow T(P')$ is an algebra homomorphism and sends $\overline{Q}$ into $\overline{Q'}$. Also, since $K(P)$ and Z are defined by algebra words, $T(\sigma)$ sends $K(P)$ into $K(P')$ and Z into Z' . Hence σ induces maps $B(M) \rightarrow B'(M)$ and $C(M) \rightarrow C'(M)$ (in the obvious notation). Similarly, there is a map $\tau : P' \rightarrow P$ inducing maps in the opposite direction. It is enough, therefore, to show that if $\rho : P \rightarrow P$ induces the identity map on M then it induces the identity map on $B(M)$ and on $C(M)$. But this is clear since if $u \in K(P)$ then u is a sum of elements $w_i(\mathbf{p})$ ($p_j \in P$), and so $u\rho - u = \sum (w_i(\mathbf{p}\rho) - w_i(\mathbf{p})) \in Z$ (because $p_j\rho - p_j \in Q$). A similar argument with 1_M replaced by an arbitrary map of R -modules shows that B and C are functors from R -modules to R -modules⁹.

We now show that $C(M) = K(P)/Z$ is a Lie structure over M in a natural way. We have already shown that $C(M)$ is a $T(M)$ -bimodule. We also know that $K(P)$ is a Lie structure over P with respect to the operation $[x, y] = xy - yx$ ($x, y \in P$). Suppose that $x \equiv x' \pmod{Q}$ and $y \equiv y' \pmod{Q}$. Then $[x, y] \equiv [x', y'] \pmod{Z}$, so for $\xi, \eta \in M$ we may define $[\xi, \eta] \in C(M) = K(P)/Z$ to be the image in $C(M)$ of $[x, y]$, where $x, y \in P$ have images ξ, η in M . The axioms for a Lie structure hold in $C(M)$ over M because they hold in $K(P)$ over P .

Theorem 4. *For any R -module M , $(C(M), [\cdot, \cdot])$ is the universal Lie structure over M .*

Proof. Let $(A, \langle \cdot, \cdot \rangle)$ be any Lie structure over M . In constructing $C(M)$ we may choose P to be a free R -module, and in this case we know (Theorem 3, Corollary 1) that $(K(P), [\cdot, \cdot])$ is the universal Lie structure over P . Now A can be viewed as a Lie structure over P *via* the map

⁹*Comment by DG:* Let me expand upon this somewhat. Let me show that B is a functor (the proof for C is analogous). Each R -module M has a canonical presentation $0 \rightarrow Q_M \rightarrow P_M \rightarrow M \rightarrow 0$, where P_M is the free R -module with M as a basis, where $P_M \rightarrow M$ is the canonical R -module epimorphism that sends each basis element corresponding to some $m \in M$ to the respective m itself, and where Q_M is the kernel of this epimorphism. We can define $B(M)$ canonically (not just up to isomorphism) using this presentation. To show that this gives a functor, we define $B(f) : B(M) \rightarrow B(N)$ for any R -linear map $f : M \rightarrow N$ as follows: Lift f to an R -linear map $P_f : P_M \rightarrow P_N$ (this lift is not canonical, but exists because P_M is free and thus projective). Then, P_f sends Q_M to Q_N , and thus we obtain a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & Q_M & \longrightarrow & P_M & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow Q_f & & \downarrow P_f & & \downarrow f & & \\ 0 & \longrightarrow & Q_N & \longrightarrow & P_N & \longrightarrow & N & \longrightarrow & 0. \end{array}$$

This leads to an algebra homomorphism $T(P_f) : T(P_M) \rightarrow T(P_N)$ that sends $K(P_M)$ to $K(P_N)$, sends Z_M to Z_N , and sends $\overline{Q}_M$ to $\overline{Q}_N$, and thus gives rise to an R -linear map $B(f) : B(M) \rightarrow B(N)$. It remains to show that this $B(f)$ depends only on f and not on its lift P_f (functoriality will then follow immediately, since $T(P_f)$ is functorial in P_f). But this is clear since, if P_{f1} and P_{f2} are two candidates for the lift P_f , then each element $u \in K(P_M)$ is a sum of elements $w_i(\mathbf{p})$ with $p_j \in P_M$, and so $uT(P_{f1}) - uT(P_{f2}) = \sum (w_i(\mathbf{p}P_{f1}) - w_i(\mathbf{p}P_{f2})) \in Z$ (because $p_jP_{f1} - p_jP_{f2} \in Q_N$).

$\theta : P \rightarrow M$ of the presentation, so there is a unique map $\alpha : K(P) \rightarrow A$ of $T(P)$ -bimodules sending $[x, y]$ to $\langle x\theta, y\theta \rangle$ for all $x, y \in P$. If $x_1, x_2, \dots, x_n \in P$ then

$$\begin{aligned} w_i(x_1, x_2, \dots, x_n)\alpha &= (x_1 \cdots x_{i-1}[x_i, x_{i+1}]x_{i+2} \cdots x_n)\alpha \\ &= (x_1\theta) \cdots (x_{i-1}\theta)\langle x_i\theta, x_{i+1}\theta \rangle(x_{i+2}\theta) \cdots (x_n\theta). \end{aligned}$$

Hence, for $q_1, q_2, \dots, q_n \in Q$, $(w_i(\mathbf{x} + \mathbf{q}) - w_i(\mathbf{x}))\alpha = 0$, i.e. $Z \subset \text{Ker } \alpha$. Thus α induces a map $\beta : C(M) = K(P)/Z \rightarrow A$ ¹⁰ sending $[\xi, \eta]$ to $\langle \xi, \eta \rangle$ for all $\xi, \eta \in M$. It is easy to see that β is a morphism of $T(M)$ -bimodules, and it is unique since the $[x, y]$ generate $K(P)$ as $T(P)$ -bimodule and therefore the $[\xi, \eta]$ generate $C(M)$ as $T(M)$ -bimodule. $\square$

The canonical maps $C(M) \rightarrow L(M)$ given by this theorem form a natural equivalence of functors $C \simeq L$. We may therefore write $L(M)$ for $C(M)$ from now on, and we have the exact sequence

$$0 \rightarrow B(M) \rightarrow L(M) \rightarrow K(M) \rightarrow 0 \quad (6')$$

for all R -modules M . Clearly this gives an exact sequence of functors $0 \rightarrow B \rightarrow L \rightarrow K \rightarrow 0$, and combining it with $0 \rightarrow K \rightarrow T \rightarrow S \rightarrow 0$ we obtain the exact sequence of functors

$$0 \rightarrow B \rightarrow L \rightarrow T \rightarrow S \rightarrow 0. \quad (7)$$

Theorem 5. *For any R -module M the following are equivalent:*

- (i) $B(M) = 0$;
- (ii) $(K(M), [,])$ is the universal Lie structure over M ;
- (iii) every Lie structure over M is embeddable in an associative structure over M .

Proof. The equivalence of (i) and (ii) follows from the exact sequence (6') (which is a consequence of Theorem 4).

(ii) $\Rightarrow$ (iii). Let $(A, \langle, \rangle)$ be any Lie structure over M and let A_0 be the $T(M)$ -bimodule generated in A by the elements $\langle x, y \rangle$ ($x, y \in M$). If (ii) holds then there is a unique morphism $\alpha : K(M) \rightarrow A$ of $T(M)$ -bimodules sending $[x, y]$ to $\langle x, y \rangle$. The kernel D of α is a $T(M)$ -bimodule, i.e. an ideal of $T(M)$. The algebra $A^* = T(M)/D$ is an associative structure over M and the Lie structure $A_0 \cong K(M)/D$ is embedded in it. To extend this to an embedding of A itself is a trivial matter. Any $T(M)$ -bimodule containing A^* is also an associative structure over M , so we need only form the fibre coproduct of A and A^* with respect to the embeddings $A_0 \rightarrow A$ and $A_0 \rightarrow A^*$. It is clear that $\langle x, y \rangle$ goes to $xy - yx$ in the resulting embedding of A .

(iii) $\Rightarrow$ (ii). If (iii) holds then the universal Lie structure $L = (L(M), \langle, \rangle)$ is embeddable in an associative structure $(L^*, (,))$ so that $(x, y) - (y, x) = \langle x, y \rangle$. By Theorem 1, there is a morphism of $T(M)$ -bimodules $\alpha : M^2T(M) \rightarrow L^*$ sending xy to (x, y) , and this induces a morphism $\beta : K(M) \rightarrow L$ sending $xy - yx$ to $\langle x, y \rangle$. Clearly β is inverse to the canonical map $L(M) \rightarrow K(M)$, so $K(M) \cong L(M)$. $\square$

¹⁰Comment by DG: The original says " $\beta : C(M) = K(M)/Z \rightarrow A$ ". Claude corrected this autonomously.

6. Modules with $B(M) = 0$

Our main result is an immediate consequence of Theorems 2 and 5:

Theorem 6. *If $B(M) = 0$ for the R -module M then the Birkhoff–Witt theorem holds for all Lie algebras over R with underlying module M .*

To show that this theorem contains the known results quoted in the introduction we now look for conditions on the R -module M which ensure that $B(M) = 0$.

Theorem 7. *Let R be a fixed commutative ring and let M be any R -module.*

- (i) *If M is R -projective then $B(M) = 0$.*
- (ii) *If M is uniquely divisible as Abelian group (i.e. M is a rational vector space) then $B(M) = 0$.*
- (iii) *If M is a direct sum of cyclic (i.e. one-generator) modules then $B(M) = 0$.*

Proof. Let $0 \rightarrow Q \rightarrow P \rightarrow M \rightarrow 0$ be a presentation of M with P projective. Then, in the notation of Section 5, $B(M) = (K(P) \cap \overline{Q})/Z$. Item (i) is clear since if M is projective we may take $P = M$ and $Q = 0$. To prove (ii) and (iii) we first observe that $K(P)$, $\overline{Q}$ and Z are homogeneous ideals of $T(P)$, so it is enough to take $u \in K(P) \cap \overline{Q}$ homogeneous of degree n ($n \geq 2$) and show that $u \in Z$. Now the symmetric group $\mathcal{S}_n$ acts on the homogeneous part $\otimes^n P$ of $T(P)$, and if $\pi \in \mathcal{S}_n$, $u \in \otimes^n P$, then $u - u\pi \in K(P)$. Moreover, if $u \in \overline{Q}$, then $u - u\pi$ is a sum of elements of type $p_1 p_2 \cdots p_{i-1} [p_i, q] p_{i+2} \cdots p_n$ or $p_1 \cdots p_{i-1} [p_i, p_{i+1}] p_{i+2} \cdots q \cdots p_n$, where $p_j \in P$ and $q \in Q$. All such elements are in Z , by definition, so $u - u\pi \in Z$ whenever $u \in \overline{Q} \cap \otimes^n P$. On the other hand, if $u \in K(P)$ and is homogeneous of degree n then u is a sum of elements of the form $v - v\tau$, where $\tau \in \mathcal{S}_n$ is a transposition. Hence $\sum_{\pi \in \mathcal{S}_n} u\pi = \sum_{\pi \in \mathcal{S}_n} (v\pi - v\tau\pi) = 0$ in this case. Thus, for any $u \in \overline{Q} \cap K(P) \cap \otimes^n P$, we have $n!u = \sum_{\pi \in \mathcal{S}_n} (u - u\pi) \in Z$. This shows that $B(M)$ is always a torsion group. It is graded by degree: $B(M) = \bigoplus B^n(M)$, and $n!B^n(M) = 0$.

Suppose now that M is uniquely divisible. Then for each integer $k > 0$ we have an isomorphism $k : M \rightarrow M$ ($x \mapsto kx$). Since B is a functor this induces an isomorphism $B(k) : B(M) \rightarrow B(M)$ which in dimension n is multiplication by k^n . Taking $k = n!$ we see that $B^n(M) = (n!)^n B^n(M) = 0$ ¹¹, which proves (ii).

To prove (iii) we suppose that $M = \bigoplus_{x \in X} M_x$, where $M_x = R/I_x$ is cyclic, and we take P to be the free R -module on X with the obvious map $P \rightarrow M$. Then Q is spanned by certain elements of the form λx , where $\lambda \in R$ and $x \in X$. We take a fixed total ordering $\leq$ of X and denote by S^* the R -submodule of $T(P)$ spanned by all products $x_1 x_2 \cdots x_n$ with $x_i \in X$ and $x_1 \leq x_2 \leq \cdots \leq x_n$ ($n \geq 0$). Then there is an R -linear map $\theta : T(P) \rightarrow S^*$ which sends any product of x 's to¹² the product of the same x 's in correct order. The kernel of θ is exactly $K(P)$. Suppose now that $u \in \overline{Q} \cap \otimes^n P$. Because of the special form of our presentation we have $u = u_1 + u_2 + \cdots + u_k$ where each u_i is still in $\overline{Q}$ and is of the form $\lambda x_1 x_2 \cdots x_n$ with $\lambda \in R$ and $x_1, x_2, \dots, x_n \in X$. Then $u_i \theta = u_i \pi_i$ for some $\pi_i \in \mathcal{S}_n$, so $u_i - u_i \theta = u_i - u_i \pi_i \in Z$,

¹¹ *Comment by DG:* Claude fixed a typo here (" $B^n(m)$ ").

¹² *Comment by DG:* In the original "to be".

as we have already shown. Hence $u - u\theta \in Z$. If now $u \in \overline{Q} \cap K(P) \cap \otimes^n P$ then $u\theta = 0$, and we have $u \in Z$ as required. $\square$

Corollary. *If R is the direct sum of a finite number of fields or is an algebra over the rationals then $B(M) = 0$ for all R -modules M . If R is a principal ideal domain then $B(M) = 0$ for all finitely generated R -modules M .*

We can extend this last result by general arguments as follows.

Theorem 8. *If $\{M_\alpha\}$ is a directed system of R -modules, and $M = \varinjlim M_\alpha$, then $B(M) = \varinjlim B(M_\alpha)$.*

Proof. The exact sequence of functors (7) gives rise to a directed system of exact sequences

$$0 \rightarrow B(M_\alpha) \rightarrow L(M_\alpha) \rightarrow T(M_\alpha) \rightarrow S(M_\alpha) \rightarrow 0.$$

Since $\varinjlim$ is an exact functor for R -modules, we obtain a commutative diagram

$$\begin{array}{ccccccccc} 0 & \rightarrow & \varinjlim B(M_\alpha) & \rightarrow & \varinjlim L(M_\alpha) & \rightarrow & \varinjlim T(M_\alpha) & \rightarrow & \varinjlim S(M_\alpha) & \rightarrow & 0 \\ & & \downarrow & & \downarrow \lambda & & \downarrow \tau & & \downarrow \sigma & & \\ 0 & \longrightarrow & B(M) & \longrightarrow & L(M) & \longrightarrow & T(M) & \longrightarrow & S(M) & \longrightarrow & 0 \end{array}$$

with exact rows, the limits in the upper row being taken in the category of R -modules. It is enough, therefore, to show that λ , τ , σ are isomorphisms. We give the proof for λ ; the other cases are proved by similar “general nonsense” and are in any case well known.

$L_\alpha = L(M_\alpha)$ is the universal Lie structure over M_α . Its structure is given by canonical maps $M_\alpha \otimes M_\alpha \rightarrow L_\alpha$, $L_\alpha \otimes M_\alpha \rightarrow L_\alpha$ and $M_\alpha \otimes L_\alpha \rightarrow L_\alpha$ satisfying axioms (L1), (L2), (L3). If $\theta : M_\alpha \rightarrow M_\beta$ is R -linear then $\theta^* = L(\theta) : L_\alpha \rightarrow L_\beta$ is obtained by viewing L_β as a Lie structure over M_α via the map θ . It is therefore not only R -linear but is compatible with the structure maps, that is, for $x, y \in M_\alpha$ and $a \in L_\alpha$ we have¹³ $\langle x, y \rangle \theta^* = \langle x\theta, y\theta \rangle$, $(ax)\theta^* = (a\theta^*)(x\theta)$ and $(xa)\theta^* = (x\theta)(a\theta^*)$. Hence, writing $\Lambda = \varinjlim L_\alpha$, the structure maps of the various L_α induce maps

$$M \otimes M = \varinjlim (M_\alpha \otimes M_\alpha) \rightarrow \Lambda, \quad \Lambda \otimes M = \varinjlim (L_\alpha \otimes M_\alpha) \rightarrow \Lambda$$

and $M \otimes \Lambda \rightarrow \Lambda$. The axioms (L1), (L2), (L3) carry over to the limits since each axiom involves only a finite number of symbols and therefore each instance of it is implied by the corresponding axiom for some pair M_α, L_α . Thus Λ is in a canonical way a Lie structure over M . If now A is any Lie structure over M then A can be viewed as a Lie structure over M_α . Hence there is a unique morphism of Lie structures $L_\alpha \rightarrow A$ for each α . These induce a unique morphism $\Lambda \rightarrow A$ of Lie structures over M which, in the particular case $A = L(M)$, is the map λ . The standard argument for universal objects now shows that λ is an isomorphism. $\square$

Corollary. *If R is a principal ideal domain then $B(M) = 0$ for all R -modules M .*

¹³Comment by DG: Missing subscripts inserted in “ $x, y \in M$ and $a \in L$ ”.

Finally, we consider change-of-ring arguments. If R' is a commutative R -algebra and M is an R -module then $M' = M \otimes_R R'$ is an R' -module and we may form its Baer invariant as such. We write $B_{R'}(M')$ to indicate that we are calculating with R' -modules.

Theorem 9. *If the R -algebra R' is flat over R then $B_{R'}(M \otimes_R R') = B_R(M) \otimes_R R'$.*

Proof. The argument is similar to the one given for direct limits. Since R' is flat over R we have, for any R -module M , an exact sequence of R' -modules

$$0 \rightarrow B_R(M) \otimes_R R' \rightarrow L_R(M) \otimes_R R' \rightarrow T_R(M) \otimes_R R' \rightarrow S_R(M) \otimes_R R' \rightarrow 0.$$

We write $M' = M \otimes_R R'$, $L = L_R(M)$, $L' = L_R(M) \otimes_R R'$. The structure maps for L , namely the maps¹⁴ $M \otimes_R M \rightarrow L$, $L \otimes_R M \rightarrow L$, $M \otimes_R L \rightarrow L$, induce R' -linear maps $M' \otimes_{R'} M' \rightarrow L'$ etc. which clearly make L' a Lie structure over M' , and it is easy to check that L' is then the universal Lie structure $L_{R'}(M')$. Similarly, we may identify $T_R(M) \otimes_R R'$ with $T_{R'}(M')$ and $S_R(M) \otimes_R R'$ with $S_{R'}(M')$, and the theorem follows. $\square$

In particular, the local ring $R_{\mathfrak{p}}$ at a prime ideal $\mathfrak{p}$ of R is flat over R (see, for example, Nagata [6], (6.18) on¹⁵ p. 19). Writing $M_{\mathfrak{p}}$ for $M \otimes_R R_{\mathfrak{p}}$ we therefore have the following.

Corollary 1. *For any R -module M and any prime ideal $\mathfrak{p}$ of R , $B_{R_{\mathfrak{p}}}(M_{\mathfrak{p}}) = (B_R(M))_{\mathfrak{p}}$. Hence $B_R(M) = 0$ if and only if $B_{R_{\mathfrak{p}}}(M_{\mathfrak{p}}) = 0$ for all prime ideals (or all maximal ideals) $\mathfrak{p}$.*

Since the local rings of a Dedekind domain are principal ideal domains, the corollary to Theorem 8 now gives

Corollary 2. *If R is a Dedekind domain then $B(M) = 0$ for all R -modules M .*

Appendix 1 (by DG). A cleaner proof of equation (3)

This appendix gives a less index-heavy proof of equation (3) from Section 4. This proof was suggested by GPT-5.5, and rewritten by myself in a more elementary language. It was originally inspired by the theory of rewrite systems [8, 11].

We keep the notation from the proof of Theorem 3. Thus X is a totally ordered basis of the free R -module M , and we have defined elements $x \circ \sigma$ and $\sigma \circ y$ of A for all $x, y \in X$ and all basis monomials $\sigma = \xi_1 \xi_2 \cdots \xi_n$ of $S(M)$.

A *word* shall mean a finite list of elements of X . We shall write a word $(w_1, w_2, \dots, w_n)$ as $w_1 w_2 \cdots w_n$, thus visually identifying it with the basis element of $T(M)$ it corresponds to.

The *inversions* of a word $w_1 w_2 \cdots w_n$ are the pairs (i, j) of integers satisfying $1 \leq i < j \leq n$ and $w_i > w_j$. In other words, they are the pairs of positions in the word such that the earlier position contains a (strictly) larger letter than the later one. We let $\text{inv } w$ denote the number of inversions of a word $w = w_1 w_2 \cdots w_n$. This is a nonnegative integer.

¹⁴ *Comment by DG:* Added “namely the maps” to avoid a false impression that had me confused too.

¹⁵ *Comment by DG:* Added “(6.18) on” to avoid relying on page numbers.

If $w = w_1 w_2 \cdots w_n$ is a word and $1 \leq i < n$ an integer satisfying $w_i > w_{i+1}$, then we can swap the i -th and $(i+1)$ -st letters of w to obtain a new word

$$w^{\curvearrowright i} := (w_1 w_2 \cdots w_{i-1}) w_{i+1} w_i (w_{i+2} w_{i+3} \cdots w_n),$$

which has one fewer inversion than w ; that is,

$$\text{inv}(w^{\curvearrowright i}) = \text{inv } w - 1. \quad (8)$$

We shall say that $w^{\curvearrowright i}$ is obtained from w by a *single sorting step* in this case. Often, several different single sorting steps can be applied to a given word w ; for example, from the word $w = 4231$ (over the totally ordered set $X = \mathbb{Z}$), we can obtain either $w^{\curvearrowright 1} = 2431$ or $w^{\curvearrowright 3} = 4213$ by a single sorting step.

Now, for any word $w = w_1 w_2 \cdots w_n$, we want to define recursively an element $J(w) \in A$ as follows:

1. If $\text{inv } w = 0$ (that is, if w has no inversions), then we set $J(w) = 0$ in A .
2. If $\text{inv } w > 0$, then there exists some integer $1 \leq i < n$ satisfying $w_i > w_{i+1}$ (since otherwise, we would have $w_1 \leq w_2 \leq \cdots \leq w_n$ and thus $\text{inv } w = 0$). We choose such an i and set

$$J(w) = w_1 w_2 \cdots w_{i-1} \langle w_i, w_{i+1} \rangle w_{i+2} w_{i+3} \cdots w_n + J(w^{\curvearrowright i}). \quad (9)$$

where (as we recall) $w^{\curvearrowright i} = (w_1 w_2 \cdots w_{i-1}) w_{i+1} w_i (w_{i+2} w_{i+3} \cdots w_n)$ is the word obtained from w by a single sorting step swapping the i -th and $(i+1)$ -st letters.

It is not obvious that this is well-defined, because in the $\text{inv } w > 0$ case, there might be several different integers $1 \leq i < n$ satisfying $w_i > w_{i+1}$, and then the formula (9) might perhaps give different results depending on which of these i is chosen. We shall soon see that all choices of i lead to the same $J(w)$ – so that $J(w)$ really is well-defined – but this must be proved. What is clear is that there is always at least one candidate for $J(w)$, because the recursion is well-founded (thanks to (8)).

We now want to show that all choices of i do lead to the same $J(w)$ in (9). For now, let us modify the definition so that the equation (9) is used to define $J(w)$ only when i is the **smallest** integer $1 \leq i < n$ satisfying $w_i > w_{i+1}$. This way, the uniqueness of $J(w)$ is obvious (since the smallest i is always unique). But we must now prove that (9) holds for **each** integer $1 \leq i < n$ satisfying $w_i > w_{i+1}$, not just for the smallest.

Proof of (9). We shall prove this by strong induction on $\text{inv } w$. The *base case* ($\text{inv } w = 0$) is obvious, since (9) does not apply to this case at all. For the *induction step*, we pick some positive integer k , and we assume that (9) is proved for all words w with $\text{inv } w < k$ (and all i that satisfy $w_i > w_{i+1}$, not just for the smallest such i). Now we shall prove the same for all words w with $\text{inv } w = k$.

Fix such a word $w = w_1 w_2 \cdots w_n$. Let j be the **smallest** integer $1 \leq i < n$ satisfying $w_i > w_{i+1}$. Then, by our (modified) definition of $J(w)$, we have

$$J(w) = w_1 w_2 \cdots w_{j-1} \langle w_j, w_{j+1} \rangle w_{j+2} w_{j+3} \cdots w_n + J(w^{\curvearrowright j}). \quad (10)$$

Now, pick **any** integer $1 \leq i < n$ satisfying $w_i > w_{i+1}$. We must prove (9). If $i = j$, then this follows from (10); thus, we assume that $i \neq j$ from now on. Since j was the smallest i satisfying

$w_i > w_{i+1}$, we have $i \geq j$, and thus $i > j$ (since $i \neq j$). Hence, we are in one of the following two cases:

Case 1: We have $i = j + 1$.

Case 2: We have $i > j + 1$.

Let us first consider Case 1. In this case, $i = j + 1$. Thus, $w_i > w_{i+1}$ (which holds by assumption) rewrites as $w_{j+1} > w_{j+2}$. Moreover, the definition of j shows that $w_j > w_{j+1}$. Therefore, $w_j > w_{j+1} > w_{j+2}$. We shall use the notations $a := w_j$, $b := w_{j+1}$, $c := w_{j+2}$, $\ell := w_1 w_2 \cdots w_{j-1}$ and $r := w_{j+3} w_{j+4} \cdots w_n$; thus, $w = labcr$ (since $w = w_1 w_2 \cdots w_n$) and $a > b > c$ (since $w_j > w_{j+1} > w_{j+2}$).

We have $w_1 w_2 \cdots w_{j-1} = \ell$ and $w_j = a$ and $w_{j+1} = b$ and $w_{j+2} w_{j+3} \cdots w_n = cr$ and therefore $w^{\curvearrowright j} = lbacr$. Thus, the equality (10) rewrites as

$$J(w) = \ell \langle a, b \rangle cr + J(lbacr). \quad (11)$$

Moreover, the word $lbacr$ is obtained from $labcr = w$ by a single sorting step (since $a > b$); thus, (8) yields $\text{inv}(lbacr) = \text{inv } w - 1 < \text{inv } w = k$. Hence, by the induction hypothesis, (9) holds for $lbacr$ instead of w . In particular, we can apply (9) to $lbacr$ and $j + 1$ instead of w and i (since the $(j + 1)$ -st and $(j + 2)$ -nd letters of $lbacr$ are a and c , and these satisfy $a > c$), and thus we obtain

$$J(lbacr) = \ell b \langle a, c \rangle r + J(lbcar). \quad (12)$$

Furthermore, the word $lbcar$ is obtained from $lbacr$ by a single sorting step (since $a > c$); thus, (8) yields $\text{inv}(lbcar) = \text{inv}(lbacr) - 1 < \text{inv}(lbacr) < k$. Hence, by the induction hypothesis, (9) holds for $lbcar$ instead of w . In particular, we can apply (9) to $lbcar$ and j instead of w and i (since the j -th and $(j + 1)$ -st letters of $lbcar$ are b and c , and these satisfy $b > c$), and thus we obtain

$$J(lbcar) = \ell \langle b, c \rangle ar + J(lcbar). \quad (13)$$

Now, (11) becomes

$$\begin{aligned} J(w) &= \ell \langle a, b \rangle cr + J(lbacr) = \ell \langle a, b \rangle cr + \ell b \langle a, c \rangle r + J(lbcar) && \text{(by (12))} \\ &= \ell \langle a, b \rangle cr + \ell b \langle a, c \rangle r + \ell \langle b, c \rangle ar + J(lcbar) && \text{(by (13))} \\ &= \ell (\langle a, b \rangle c + b \langle a, c \rangle + \langle b, c \rangle a) r + J(lcbar). \end{aligned} \quad (14)$$

On the other hand, our goal is to prove (9). In other words, our goal is to prove

$$J(w) = \ell a \langle b, c \rangle r + J(lacbr) \quad (15)$$

(since $i = j + 1$, and thus $w_1 w_2 \cdots w_{i-1} = w_1 w_2 \cdots w_j = \ell a$ and $w_i = w_{j+1} = b$ and $w_{i+1} = w_{j+2} = c$ and $w_{i+2} w_{i+3} \cdots w_n = w_{j+3} w_{j+4} \cdots w_n = r$ and $w^{\curvearrowright i} = lacbr$). Let us rewrite the right hand side.

The word $lacbr$ is obtained from $labcr = w$ by a single sorting step (since $b > c$); thus, (8) yields $\text{inv}(lacbr) = \text{inv } w - 1 < \text{inv } w = k$. Hence, by the induction hypothesis, (9) holds for $lacbr$ instead of w . In particular, we can apply (9) to $lacbr$ and j instead of w and i (since the j -th and $(j + 1)$ -st letters of $lacbr$ are a and c , and these satisfy $a > c$), and thus we obtain

$$J(lacbr) = \ell \langle a, c \rangle br + J(lcabr). \quad (16)$$

The word $lcabr$ is obtained from $lacbr$ by a single sorting step (since $a > c$); thus, (8) yields $\text{inv}(lcabr) = \text{inv}(lacbr) - 1 < \text{inv}(lacbr) < k$. Hence, by the induction hypothesis, (9) holds for

$\ell cabr$ instead of w . In particular, we can apply (9) to $\ell cabr$ and $j+1$ instead of w and i (since the $(j+1)$ -st and $(j+2)$ -nd letters of $\ell cabr$ are a and b , and these satisfy $a > b$), and thus we obtain

$$J(\ell cabr) = \ell c \langle a, b \rangle r + J(\ell cbar).$$

Substituting this into (16), we find

$$J(\ell acbr) = \ell \langle a, c \rangle br + \ell c \langle a, b \rangle r + J(\ell cbar). \quad (17)$$

However, the axiom (L3) of the Lie structure A yields

$$(\langle a, b \rangle c - c \langle a, b \rangle) + (\langle b, c \rangle a - a \langle b, c \rangle) + (\langle c, a \rangle b - b \langle c, a \rangle) = 0.$$

In view of $\langle c, a \rangle = -\langle a, c \rangle$ (this is an easy consequence of axiom (L1), just as for Lie algebras), we can rewrite this as

$$(\langle a, b \rangle c - c \langle a, b \rangle) + (\langle b, c \rangle a - a \langle b, c \rangle) + ((-\langle a, c \rangle) b - b(-\langle a, c \rangle)) = 0.$$

Expanding the parentheses and collecting the negative terms on the right hand side, we transform this into

$$\langle a, b \rangle c + \langle b, c \rangle a + b \langle a, c \rangle = c \langle a, b \rangle + a \langle b, c \rangle + \langle a, c \rangle b. \quad (18)$$

Now, (14) becomes

$$\begin{aligned} J(w) &= \ell (\underbrace{\langle a, b \rangle c + b \langle a, c \rangle + \langle b, c \rangle a}_{\substack{= \langle a, b \rangle c + \langle b, c \rangle a + b \langle a, c \rangle \\ = c \langle a, b \rangle + a \langle b, c \rangle + \langle a, c \rangle b \\ \text{(by (18))}}}) r + J(\ell cbar) \\ &= \ell (c \langle a, b \rangle + a \langle b, c \rangle + \langle a, c \rangle b) r + J(\ell cbar) \\ &= \ell c \langle a, b \rangle r + \ell a \langle b, c \rangle r + \ell \langle a, c \rangle br + J(\ell cbar) \\ &= \ell a \langle b, c \rangle r + \ell \langle a, c \rangle br + \underbrace{\ell c \langle a, b \rangle r + J(\ell cbar)}_{\substack{= J(\ell acbr) \\ \text{(by (17))}}} \\ &= \ell a \langle b, c \rangle r + J(\ell acbr). \end{aligned}$$

This proves (15). In other words, (9) is proved for our arbitrary i (since (15) is just a rewritten form of (9)). This completes the induction step in Case 1.

Let us now consider Case 2. In this case, $i > j+1$. Thus, the four integers $j, j+1, i, i+1$ are distinct and ordered as follows: $j < j+1 < i < i+1$. We shall use the notations $a := w_j$, $b := w_{j+1}$, $c := w_i$, $d := w_{i+1}$, $\ell := w_1 w_2 \cdots w_{j-1}$, $m := w_{j+2} w_{j+3} \cdots w_{i-1}$ and $r := w_{i+2} w_{i+3} \cdots w_n$; thus, $w = \ell b m c d r$ (since $w = w_1 w_2 \cdots w_n$ and $j < j+1 < i < i+1$) and $a > b$ (since $w_j > w_{j+1}$) and $c > d$ (since $w_i > w_{i+1}$).

We have $w_1 w_2 \cdots w_{j-1} = \ell$ and $w_j = a$ and $w_{j+1} = b$ and $w_{j+2} w_{j+3} \cdots w_n = m c d r$ and thus $w^{\sim j} = \ell b a m c d r$. Thus, the equality (10) rewrites as

$$J(w) = \ell \langle a, b \rangle m c d r + J(\ell b a m c d r). \quad (19)$$

Moreover, the word $\ell b a m c d r$ is obtained from $\ell a b m c d r = w$ by a single sorting step (since $a > b$); thus, (8) yields $\text{inv}(\ell b a m c d r) = \text{inv} w - 1 < \text{inv} w = k$. Hence, by the induction hypothesis, (9) holds for $\ell b a m c d r$ instead of w . In particular, we can apply (9) to $\ell b a m c d r$ and i instead of w and

i (since the i -th and $(i + 1)$ -st letters of $\ell bamcdr$ are c and d , and these satisfy $c > d$), and thus we obtain

$$J(\ell bamcdr) = \ell bam \langle c, d \rangle r + J(\ell bamdcr).$$

Substituting this into (19), we find

$$\begin{aligned} J(w) &= \ell \langle a, b \rangle mcd r + \ell bam \langle c, d \rangle r + J(\ell bamdcr) \\ &= \ell (\langle a, b \rangle mcd + bam \langle c, d \rangle) r + J(\ell bamdcr). \end{aligned} \quad (20)$$

On the other hand, our goal is to prove (9). In other words, our goal is to prove

$$J(w) = \ell abm \langle c, d \rangle r + J(\ell abmdcr) \quad (21)$$

(since $w_1 w_2 \cdots w_{i-1} = \ell abm$ and $w_i = c$ and $w_{i+1} = d$ and $w_{i+2} w_{i+3} \cdots w_n = r$ and thus $w^{\curvearrowright i} = \ell abmdcr$). Let us rewrite the right hand side.

The word $\ell abmdcr$ is obtained from $\ell bamcdr = w$ by a single sorting step (since $c > d$); thus, (8) yields $\text{inv}(\ell abmdcr) = \text{inv } w - 1 < \text{inv } w = k$. Hence, by the induction hypothesis, (9) holds for $\ell abmdcr$ instead of w . In particular, we can apply (9) to $\ell abmdcr$ and j instead of w and i (since the j -th and $(j + 1)$ -st letters of $\ell abmdcr$ are a and b , and these satisfy $a > b$), and thus we obtain

$$J(\ell abmdcr) = \ell \langle a, b \rangle mdc r + J(\ell bamdcr). \quad (22)$$

However, the axiom (L2) of the Lie structure A yields

$$\langle a, b \rangle m (cd - dc) = (ab - ba) m \langle c, d \rangle.$$

Expanding both sides, we rewrite this as

$$\langle a, b \rangle mcd - \langle a, b \rangle mdc = abm \langle c, d \rangle - bam \langle c, d \rangle.$$

Equivalently,

$$\langle a, b \rangle mcd + bam \langle c, d \rangle = abm \langle c, d \rangle + \langle a, b \rangle mdc.$$

Substituting this into (20), we obtain

$$\begin{aligned} J(w) &= \ell (abm \langle c, d \rangle + \langle a, b \rangle mdc) r + J(\ell bamdcr) \\ &= \ell abm \langle c, d \rangle r + \underbrace{\ell \langle a, b \rangle mdc r + J(\ell bamdcr)}_{=J(\ell abmdcr) \text{ (by (22))}} \\ &= \ell abm \langle c, d \rangle r + J(\ell abmdcr). \end{aligned}$$

This proves (21). In other words, (9) is proved for our arbitrary i (since (21) is just a rewritten form of (9)). This completes the induction step in Case 2.

We have now completed the induction step in both Cases 1 and 2. Thus, the induction proof of (9) is complete. $\square$

Thus, for each word $w = w_1 w_2 \cdots w_n$, we have defined an element $J(w) \in A$ that satisfies (9) for each $1 \leq i < n$ satisfying $w_i > w_{i+1}$ (not just for the smallest such i).

Next, let us define some general terminology. A word is said to be *sorted* if its letters are weakly increasing from left to right (i.e., if it has no inversions). For each word w , there is exactly one sorted word that contains the same letters (with the same multiplicities) as w ; this word is called

the *sorted rearrangement* of w , and can be obtained from w by successively applying single sorting steps until no more inversions remain. We shall use the notation $\text{sort } w$ for this word. Note that

$$J(w) = 0 \quad \text{for any sorted word } w \quad (23)$$

(by the base case of the recursive definition of $J(w)$, since $\text{inv } w = 0$).

Furthermore, for any word w (or, more generally, any element w of $T(M)$), we shall let $\overline{w}$ denote the projection of w onto $S(M)$.

We shall now prove the following general formula: If v is a word and w is a sorted word, and $x \in X$ is a letter, then

$$(x \circ \overline{w}) v = J(xwv) - J(\text{sort}(xw) v). \quad (24)$$

Proof of (24). Indeed, let v be a word, let w be a sorted word, and let $x \in X$ be a letter. Note that $\overline{\text{sort}(xw)} = \overline{xw}$ (since the words $\text{sort}(xw)$ and xw differ only in the order of their letters, and thus have the same projection onto the commutative algebra $S(M)$).

Write the sorted word w as $w = t_1 t_2 \cdots t_n$. Thus, $\overline{w} = \overline{t_1 t_2 \cdots t_n} = \tau_1 \tau_2 \cdots \tau_n$, where each τ_i is the image of t_i in $S(M)$.

Since the word $t_1 t_2 \cdots t_n$ is sorted, we have $t_1 \leq t_2 \leq \cdots \leq t_n$. Thus, we have

$$t_1 \leq t_2 \leq \cdots \leq t_{m-1} < x \leq t_m \leq t_{m+1} \leq \cdots \leq t_n \quad (25)$$

for a unique integer $1 \leq m \leq n+1$ (in particular, $m = n+1$ if $t_n < x$, whereas $m = 1$ if $x \leq t_1$). Consider this m . Thus, $t_1 t_2 \cdots t_{m-1} x t_m t_{m+1} \cdots t_n$ is a sorted word (by (25)). The definition of $x \circ \overline{w}$ yields

$$\begin{aligned} x \circ \overline{w} &= \sum_{t_i < x} t_1 t_2 \cdots t_{i-1} \langle x, t_i \rangle t_{i+1} t_{i+2} \cdots t_n \quad (\text{since } \overline{w} = \tau_1 \tau_2 \cdots \tau_n) \\ &= \sum_{i=1}^{m-1} t_1 t_2 \cdots t_{i-1} \langle x, t_i \rangle t_{i+1} t_{i+2} \cdots t_n \end{aligned} \quad (26)$$

(since the integers $i \in \{1, 2, \dots, n\}$ satisfying $t_i < x$ are precisely $1, 2, \dots, m-1$, according to (25)).

Moreover, from $w = t_1 t_2 \cdots t_n$, we obtain

$$\text{sort}(xw) = \text{sort}(x t_1 t_2 \cdots t_n) = t_1 t_2 \cdots t_{m-1} x t_m t_{m+1} \cdots t_n \quad (27)$$

(by (25)). Furthermore, the word xwv can be transformed into the word $\text{sort}(xw) v$ by the following sequence of single sorting steps:

$$\begin{aligned} xwv &= x t_1 t_2 \cdots t_n v && (\text{since } w = t_1 t_2 \cdots t_n) \\ &\mapsto t_1 x t_2 t_3 \cdots t_n v && (\text{we swapped } x \text{ with } t_1) \\ &\mapsto t_1 t_2 x t_3 t_4 \cdots t_n v && (\text{we swapped } x \text{ with } t_2) \\ &\mapsto \cdots \\ &\mapsto \underbrace{t_1 t_2 \cdots t_{m-1} x t_m t_{m+1} \cdots t_n}_{=\text{sort}(xw)} v && (\text{we swapped } x \text{ with } t_{m-1}) \\ &= \text{sort}(xw) v. \end{aligned}$$

Using the recursive definition (9) of the J -function, we thus have

$$\begin{aligned}
J(xt_1t_2 \cdots t_nv) &= \langle x, t_1 \rangle t_2t_3 \cdots t_nv + J(t_1xt_2t_3 \cdots t_nv); \\
J(t_1xt_2t_3 \cdots t_nv) &= t_1 \langle x, t_2 \rangle t_3t_4 \cdots t_nv + J(t_1t_2xt_3t_4 \cdots t_nv); \\
&\vdots; \\
J(t_1t_2 \cdots t_{m-2}xt_{m-1}t_m \cdots t_nv) &= t_1t_2 \cdots t_{m-2} \langle x, t_{m-1} \rangle t_mt_{m+1} \cdots t_nv \\
&\quad + J(t_1t_2 \cdots t_{m-1}xt_mt_{m+1} \cdots t_nv).
\end{aligned}$$

Substituting these equalities into one another, we obtain

$$\begin{aligned}
&J(xt_1t_2 \cdots t_nv) \\
&= \underbrace{\langle x, t_1 \rangle t_2t_3 \cdots t_nv + t_1 \langle x, t_2 \rangle t_3t_4 \cdots t_nv + \cdots + t_1t_2 \cdots t_{m-2} \langle x, t_{m-1} \rangle t_mt_{m+1} \cdots t_nv}_{= \sum_{i=1}^{m-1} t_1t_2 \cdots t_{i-1} \langle x, t_i \rangle t_{i+1}t_{i+2} \cdots t_nv} \\
&\quad + J\left(\underbrace{t_1t_2 \cdots t_{m-1}xt_mt_{m+1} \cdots t_nv}_{=\text{sort}(xw)}\right) \\
&= \underbrace{\sum_{i=1}^{m-1} t_1t_2 \cdots t_{i-1} \langle x, t_i \rangle t_{i+1}t_{i+2} \cdots t_nv}_{=\overline{x \circ w} \text{ (by (26))}} + J(\text{sort}(xw)v) \\
&= (x \circ \overline{w})v + J(\text{sort}(xw)v). \tag{28}
\end{aligned}$$

Since $t_1t_2 \cdots t_n = w$, this rewrites as $J(xwv) = (x \circ \overline{w})v + J(\text{sort}(xw)v)$. This proves (24). $\square$

Similarly to (24), we can prove that if u is a word and w is a sorted word, and $y \in X$ is a letter, then

$$u(\overline{w} \circ y) = J(uwy) - J(u \text{sort}(wy)). \tag{29}$$

Now, fix $x, y \in X$ and $x_1 \leq x_2 \leq \cdots \leq x_n$ in X . Let ξ, η and ξ_i be the images of x, y and x_i in $S(M)$. Let $\sigma = \xi_1\xi_2 \cdots \xi_n$. We now take aim at proving the two equalities

$$(\xi\sigma) \circ y + (x \circ \sigma)y = J(xx_1x_2 \cdots x_ny) \quad \text{and} \tag{32}$$

$$x \circ (\sigma\eta) + x(\sigma \circ y) = J(xx_1x_2 \cdots x_ny). \tag{33}$$

Once these two equalities are proved, it will immediately follow that

$$(\xi\sigma) \circ y + (x \circ \sigma)y = J(xx_1x_2 \cdots x_ny) = x \circ (\sigma\eta) + x(\sigma \circ y),$$

thus

$$(\xi\sigma) \circ y - x(\sigma \circ y) = x \circ (\sigma\eta) - (x \circ \sigma)y,$$

and thus (3) will be proved. Thus, it remains to prove (32) and (33).

The word $x_1x_2 \cdots x_n$ is sorted (since $x_1 \leq x_2 \leq \cdots \leq x_n$). Hence, applying (24) to $v = y$ and $w = x_1x_2 \cdots x_n$, we find

$$(x \circ \overline{x_1x_2 \cdots x_n})y = J(xx_1x_2 \cdots x_ny) - J(\text{sort}(xx_1x_2 \cdots x_n)y).$$

Thus,

$$\begin{aligned} J(xx_1x_2 \cdots x_ny) &= J(\text{sort}(xx_1x_2 \cdots x_n)y) + (x \circ \overline{xx_1x_2 \cdots x_n})y \\ &= J(\text{sort}(xx_1x_2 \cdots x_n)y) + (x \circ \sigma)y \end{aligned} \quad (34)$$

(since $\overline{xx_1x_2 \cdots x_n} = \xi_1\xi_2 \cdots \xi_n = \sigma$). But $\text{sort}(xx_1x_2 \cdots x_n)$ is also a sorted word. Thus, applying (29) to $u = 1$ and $w = \text{sort}(xx_1x_2 \cdots x_n)$, we find

$$1 \left(\overline{\text{sort}(xx_1x_2 \cdots x_n)} \circ y \right) = J(1 \text{ sort}(xx_1x_2 \cdots x_n)y) - J(1 \text{ sort}(\text{sort}(xx_1x_2 \cdots x_n)y)).$$

This simplifies to

$$\begin{aligned} \overline{\text{sort}(xx_1x_2 \cdots x_n)} \circ y &= J(\text{sort}(xx_1x_2 \cdots x_n)y) - \underbrace{J(\text{sort}(\text{sort}(xx_1x_2 \cdots x_n)y))}_{\substack{=0 \\ \text{(by (23), since } \text{sort}(\text{sort}(xx_1x_2 \cdots x_n)y) \\ \text{is a sorted word)}}} \\ &= J(\text{sort}(xx_1x_2 \cdots x_n)y). \end{aligned}$$

Thus,

$$J(\text{sort}(xx_1x_2 \cdots x_n)y) = \underbrace{\overline{\text{sort}(xx_1x_2 \cdots x_n)}}_{=\overline{xx_1x_2 \cdots x_n} = \xi\xi_1\xi_2 \cdots \xi_n = \xi\sigma} \circ y = (\xi\sigma) \circ y.$$

Substituting this into (34), we obtain

$$J(xx_1x_2 \cdots x_ny) = (\xi\sigma) \circ y + (x \circ \sigma)y.$$

This proves (32). A similar argument (in which we first apply (29) to $u = x$ and $w = x_1x_2 \cdots x_n$, and then apply (24) to $v = 1$ and $w = \text{sort}(x_1x_2 \cdots x_ny)$) proves (33). Hence, as we said above, (3) follows.

Appendix 2 (by DG). Birkhoff–Witt for flat modules

Theorem 7 (i) can be generalized as follows:

Theorem 10. *Let R be a fixed commutative ring, and M be a flat R -module. Then, $B(M) = 0$.*

Proof. The Govorov–Lazard theorem ([10, Théorème 1.2 (iii)], [9]) shows that M is a direct limit of a directed system $\{M_\alpha\}$ of free R -modules. By Theorem 7 (i), all these free R -modules M_α satisfy $B(M_\alpha) = 0$. Hence, Theorem 8 yields $B(M) = \varinjlim B(M_\alpha) = \varinjlim 0 = 0$. $\square$

References

- [1] BIRKHOFF, G. Representability of Lie algebras and Lie groups by matrices. *Annals Math.* **38** (1937), 526–532.
- [2] CARTIER, P. Remarques sur le théorème de Birkhoff–Witt. *Ann. Scuola norm. sup. Pisa, Sci. fis. mat.* 3 Ser. **12** (1958), 1–4.

- [3] COHN, P. M. A remark on the Birkhoff–Witt theorem. *J. London Math. Soc.* **38** (1963), 197–203.
- [4] FRÖHLICH, A. Baer-invariants of algebras. *Trans. Am. Math. Soc.* **109** (1963), 221–244.
- [5] LAZARD, M. Sur les algèbres enveloppantes universelles de certaines algèbres de Lie. *Publ. Sci. Univ. Alger, Sér. A.* **1** (1954), 281–294.
- [6] NAGATA, M. “Local rings”, (Interscience, New York, 1962).
- [7] WITT, E. Treue Darstellung Liescher Ringe. *J. reine angew. Math.* **177** (1937), 152–160. ¹⁶

Additional references inserted by DG:

- [8] BAADER, F. AND NIPKOW, T. “Term Rewriting and All That”, (Cambridge University Press, Cambridge, 1998).
- [9] HINES, R. Lazard’s theorem (characterizing flatness), notes, December 7, 2016. <https://math.colorado.edu/~rohi1040/expository/lazardstheorem.pdf>
- [10] LAZARD, D. Autour de la platitude. *Bulletin de la Société Mathématique de France* **97** (1969), 81–128.
- [11] NEWMAN, M. H. A. On theories with a combinatorial definition of “equivalence”. *Annals of Mathematics*, Second Series, **43** (1942), 223–243.

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¹⁶ *Comment by DG:* Some titles corrected.